Exp124: pause-token training for contact prediction

- Test the pause-token idea from Goyal et al. (ICLR 2024) in contacts-v1.
- Insert <think> tokens during training while masking their prediction loss.
- The full v5p-128 run completed and exported step 35680.
- Evaluate 100 rollouts per protein on the legacy 554-protein contact set.
- Conclusion: no contact-prediction improvement in this tested setting.
- README is the canonical record; committed CSVs provide the reported values.

Validation losses: compare matching loss definitions

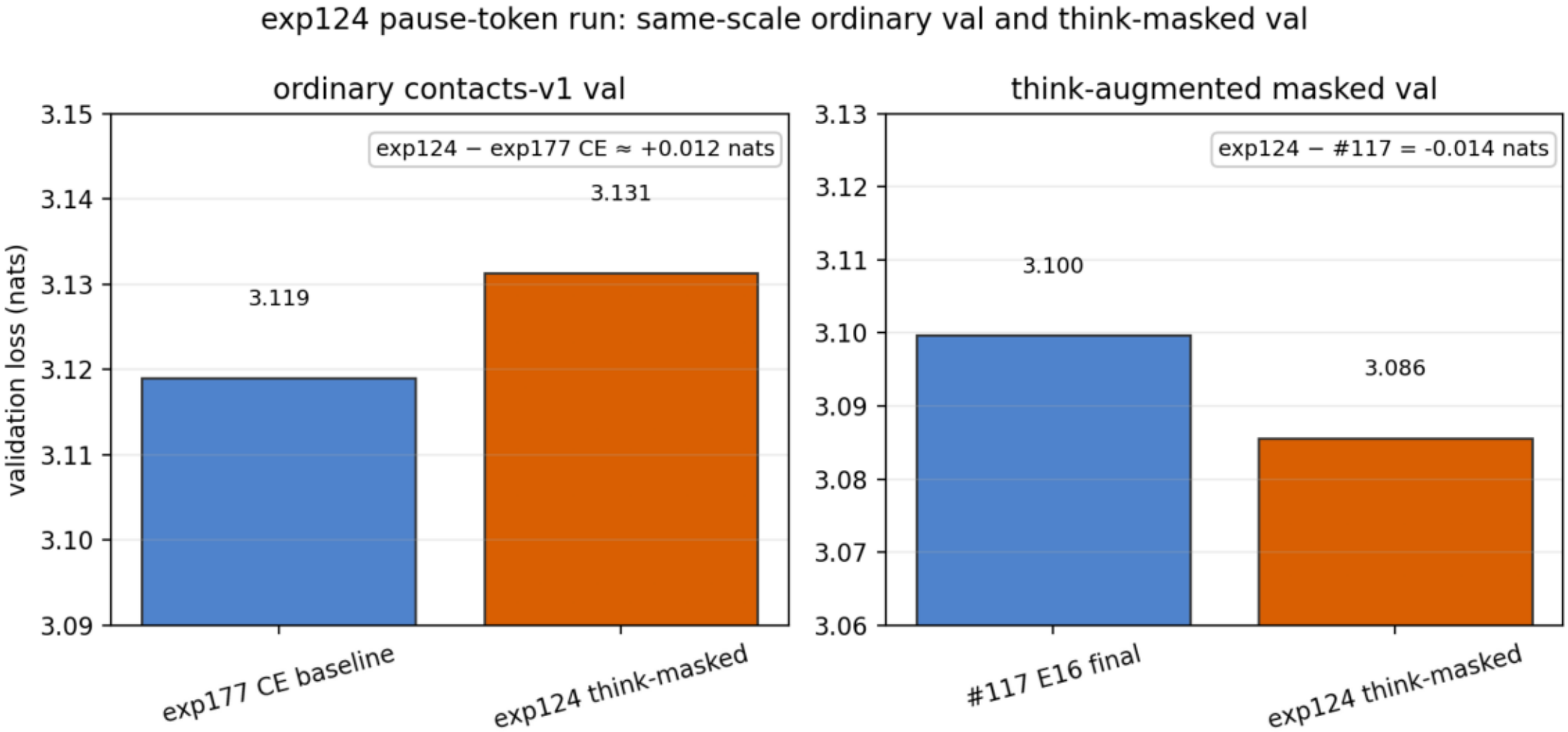
- Ordinary contacts-v1 validation: exp124 = 3.131303071975708 nats.
- Same-stack exp177 CE reference: approximately 3.119; exp124 is about 0.012 worse.
- Historical #117/#75 losses (2.7037 / 2.7566) use an older loss scale.
- Those historical values are not direct absolute-loss baselines for exp124.
- On the same think-masked cache: exp124 = 3.0855870246887207.
- Recomputed #117 on that cache = 3.099645097316767.
- The native masked-loss gain did not transfer to downstream contact accuracy.

Pause tokens do not improve contact accuracy

- Standard prompt: all R = 0.3365; long R = 0.2856; all AUC = 0.8119.
- One <think>: all R = 0.3359; long R = 0.2822; all AUC = 0.8103.
- Two <think>: all R = 0.3347; long R = 0.2805; all AUC = 0.8093.
- Three <think>: all R = 0.3353; long R = 0.2806; all AUC = 0.8093.
- Long recall@L is lower than the standard prompt in every inserted-token setting.
- This small contacts-v1 experiment did not reproduce the reported pause-token gain.

final_losses.png

Final exp124 validation losses. Ordinary contacts-v1 loss is compared to the same-era exp177 CE baseline; historical #75/#117 W&B ordinary losses use an older loss scale and are not plotted as direct baselines.




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$ plot_final_losses.py --input data/final_losses.csv --output plots/final_losses.png
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